

Scrambling for Power: The Impact of Sudden Reactor Shutdowns on Short-Run Electricity Prices

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Abstract

We study the short-run impact of unexpected nuclear reactor outages (scrams) on electricity prices by implementing a stacked Regression Discontinuity in Time (RDiT) design across multiple plants and electricity markets. Focusing on 13 nuclear scrams across 10 sites, we align each event on its own outage timestamp and estimate the instantaneous price jump per megawatt of lost capacity. According to our estimates, each 1 GW of nuclear capacity lost raises the zonal marginal price by $\$26.18 \pm 15.06/\text{MWh}$.

1 Introduction

Energy is unarguably one of the most important resources throughout human history. From the Industrial Revolution to modern conflicts in the Middle East, securing stable supplies of energy are of extreme concern for governments, corporations and consumers. Yet, over the past 50 years, nuclear energy, one of the safest [\[1\]](#) and most plentiful sources of energy has seen widespread opposition and successful efforts to remove it from the mix, primarily in Western countries. Countries ranging from Germany to Japan to the United States have all seen extensive drawdowns or stagnation in nuclear energy. Opponents of nuclear energy

claim that it is dangerous or expensive [2], while supporters tout its track record and need in a world in need of clean energy.

However, the case for nuclear energy is quite strong. First, nuclear is 'baseload' power [3]; power that is almost always on, providing a steady source of energy. This is important in providing stable supplies of energy since the growing shift to renewable energy introduces significant variability into the grid [4]. By combining these sources with a steady source like nuclear, power prices are less likely to spike (and dampened when they do) when the sun stops shining or it isn't windy. In fact, the capacity factor of nuclear energy, the fraction of time it's producing energy, is 92% [3], higher than any other energy source.

Nuclear energy is also extremely efficient, and produces massive amounts of energy for a relatively small use of resources like water and uranium. The average nuclear reactor produces about 1 GW of energy and overall nuclear produces about 20% of total U.S. energy [3].

1.1 Research Question

Given the extensive benefits of nuclear energy, and the wave of drawdowns over the past decades, quantifying the impact of these closures is of vital importance to policymakers. Dozens of nuclear reactors still exist throughout the world, and many are coming up for license renewal and repair in the next decades [5]. However, understanding the impact of closures is inherently hard because of simultaneous determination issues, as any price change in response to a supply shock will consist of some demand-shifts as well. The cyclical nature of demand also complicates this further, as prices typically jump during the day as commerce and jobs start up. Weather and seasons also complicate this as AC and heating are significant contributors to load on the grid. Much of the effect from a closure would likely be buried beneath these demand shifts. Additionally, electricity markets are fluid and interconnected, with different markets importing and exporting electricity as needed. Lastly, the self-correcting nature of markets mean that long run analyses suffer from part of

the causal effect being hidden by new entrants (new generation sources).

This study attempts to identify the causal effect of nuclear closures by exploiting sharp drops in nuclear generation in response to scrams. Scrams refer to when nuclear reactors suffer a fault and need to immediately shut down [6]. These are typically mechanical or electrical, and result in the generation of a reactor going from 100% to 0% almost instantaneously. The key attribute of these drops is the extremely short timeframes over which they occur. We can plausibly believe that normal market dynamics are relatively fixed over the short run, as it's not possible for new supply to come online in such a short span, nor for demand to adjust in response to higher prices, as typically is seen over longer timeframes.

This naturally leads to a regression discontinuity in time (RDiT) approach. We use panel data on 13 scrams from 10 nuclear reactors over four different electricity markets to construct a pooled RDiT to estimate the causal effect of nuclear generation going offline on zonal marginal prices ¹ (ZMP), prices that correspond to the electricity sub-market the plant is located in. Standard economic theory indicates that we would expect a positive *causal* effect (representing an increase in prices), but a much smaller effect on prices should we compare *equilibrium* prices, which will complicate the implementation of this strategy.

¹One electricity market, SPP, only has hub-based prices available, one of which was used instead

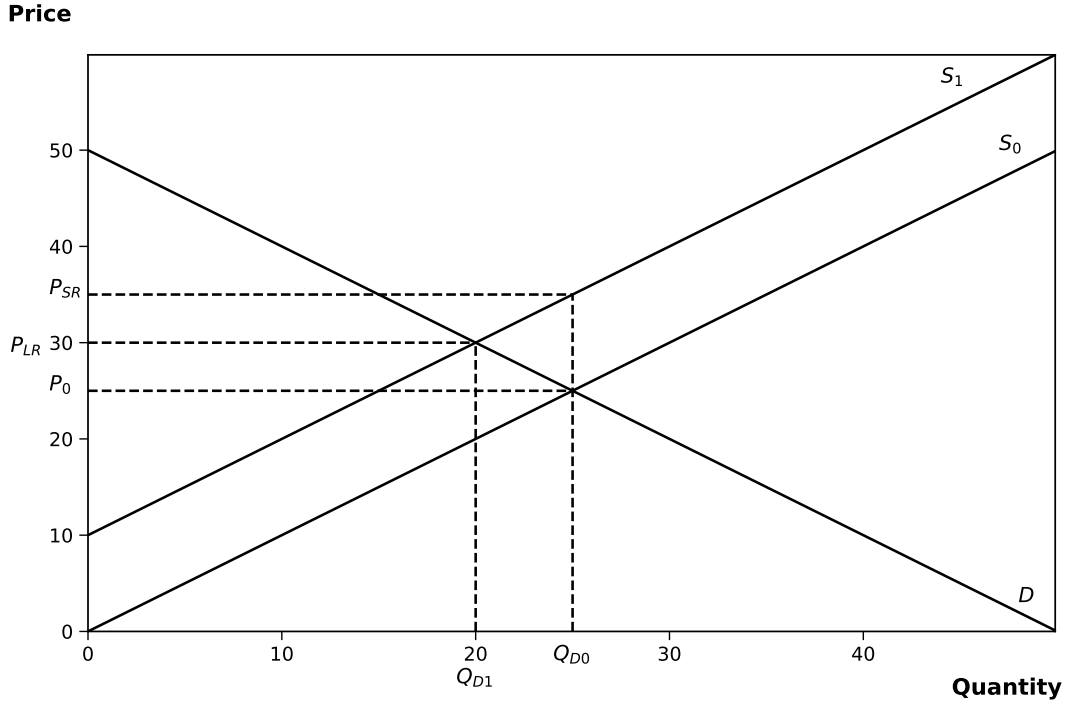


Figure 1: Stylized supply and demand graph showing price and quantity scenarios.

To see this, consider Figure 1. Our pre-scram market can be characterized by curve S_0 , D with price P_0 and quantity Q_{D0} . Once the scram occurs, we have supply characterized by curve S_1 . In the long run, we would see quantity demanded adjust and give us price P_{LR} and quantity Q_{D0} . But, if we make the assumption that in the short-run quantity demanded is fixed, we instead get price $P_{SR} > P_{LR}$. The exact magnitude of the difference will depend on the specific market conditions, but theoretically it's plausible that a long run effect is an under-estimate of the true immediate effect.

1.2 Literature Review

Studying the impact of nuclear reactor closures on electricity prices is by no measure a novel topic. Neidell, Uchida and Veronesi (2021) [7] uses an RDiT approach to study Japan's post-2011 policy of shutting all nuclear reactors within 14 months of the nuclear disaster at

Fukushima. They find that the closures caused prices to spike in some regions by nearly 40%, and caused increased mortality in colder regions. Another study, Grossi, Heim and Waterson (2014) [8], uses the same disaster at Fukushima, but instead studies the German response to it. They construct a counterfactual price trend and find that Germany’s immediate shutdown of nuclear reactors led to a 7% increase in electricity prices relative to no shutdown.

In the United States, some authors have also focused on scrams, as opposed to the previous studies which exploited a country-wide policy change. Davis and Hausman (2016) [9] study the shutdown of the San Onofre reactor using an event-study approach, and find that generation costs jumped by 13% and created zonal price imbalances. Woo et. al (2014) [10] analyzed the same shutdown and found that prices were raised by \$6-9/MWh. While these studies do exploit instantaneous shutdowns, they are all focused on long term outcomes, allowing for supply and demand to shift and discussing the new equilibrium price. These longer timespans allow for flexible supply and demand, as over months or a year, new generation can come online, and industry can adjust consumption patterns. There is also time for more consistent imports to built in, as neighboring zones and markets adjust to the shock. These estimates are still extremely important, as they provide information to policymakers on what the eventual outcome of a proposed shutdown could. Despite this, there is a need to understand the immediate short-run shock, as a way of understanding the *ceteris paribus* importance of a given nuclear plant. As discussed earlier, economic theory would tell us that estimates may be an *underestimate* of the true immediate impact of lost capacity. To that end, we know of no other papers that analyze the extremely-short run impact of outages on prices.

2 Strategy

2.1 Empirical Strategy

To implement this RDiT approach, consider the initial regression:

$$P_{it} = \beta_0 + \theta s_{it} + \rho \mathbf{1}_{scram} + \epsilon_{it}$$

This is a basic RDiT formulation, where P_{it} is the ZMP for scram i at time t , s_{it} is the time since scram i at time t (i.e. the running variable), and the indicator is whether the scram happened. Time here is at 5-minute intervals. This would give us *a* causal effect, but leaves much to be desired. Thus, consider the following regression:

$$P_{it} = \alpha_i + \lambda_{hour} + \theta_b b_{it} + \theta_a a_{it} + \rho \mathbf{1}_{scram} \times D_i + \beta_1 Load_{it} + \epsilon_{it} \quad (1)$$

We have now added a few more terms. α_i is a fixed effect at the scram level, differentiating each event. λ_{hour} is a time fixed event at the *hour of day* level, allowing for us to control for varying levels of activity during the day. b_{it} and a_{it} are the time before treatment and after treatment variables, respectively, forming the running variable here. The terms are split to allow for different locally linear pre- and post-trends. D_i is the drop size in MWh for scram i , interacted with the treatment ($\mathbf{1}_{scram}$). $Load_{it}$ is the actual load on the grid in the zone of scram i at time t .

2.2 Identification Concerns

A few issues could complicate the strategy here. First, significant other variation in generation could swamp the effect of the drop in nuclear generation. To account for this, we plotted generation over time to see if any saw wild swings during the timeframe of interest. One scram was dropped for this reason, since our assumptions regarding inelasticity of supply and demand would be violated.

Another concern could be the reactor coming back online during the treatment period, complicating the analysis since those data points could be coded as being treated. To account for this I again removed scrams that saw extremely short lived spikes downwards or with large positive jumps in power during the treatment window.

Additionally, the simultaneity concern discussed in the introduction is of credible threat to the strategy. Controlling for demand via actual load and hour of day fixed effects help mitigate this issue, but fundamentally dealing with this requires the assumption of short-run inelasticity. While we did not check this in the data more rigorously, standard economic theory provides a plausible basis for this.

A more mechanical concern could be model misspecification as the running variable (time since treatment) could have a true locally constant or quadratic relationship. We test these specifications in Figure 5.

Another concern is the choice of bandwidth. In an RDiT context, data further away from treatment is less useful, inducing bias, but using less data causes variance to substantially increase. And, given that there are repeated instances of plants and markets, there could be correlated errors across these instances. Adding fixed effects at these levels could also change the effect size. To be on the safe side, we cluster at the highest level (market) and test the various bandwidth and fixed effect specifications in Figure 5.

3 Data

3.1 Data Sourcing and Processing

The data for this study comes from two sources. The initial source was the Nuclear Regulatory Commission’s Operating Reactor Scrams Dashboard [11], which tracks every instance of a scram in the U.S. The second data source was the U.S. Energy Administration’s (EIA) Wholesale Electricity Market data [12], which contains standardized datasets on prices, generation by type and load for all regional transmissions organizations (RTOs) in the U.S. To assemble the dataset in this paper, we first limited our search to late 2020 - 2024, the range of dates covered by the EIA datasets. We focused on four RTOs ², California Independent

²Technically speaking, CAISO and NYISO are not RTOs but are Independent System Operators (ISOs). Functionally, they are the same but more broadly RTOs engage in transmission planning and capacity markets, while ISOs are less involved. RTOs are also more regulated by the federal government, due to their

System Operator (CAISO), New York Independent System Operator (NYISO), Southwest Power Pool (SPP) ³ and Pennsylvania-New Jersey-Maryland Interconnection (PJM). Figure 2 shows the geographical coverage of these markets. We picked these four RTOs as EIA had granular (5 minute intervals) price data, as well as generation by source and load data. Beyond a data perspective, these four RTOs form a substantial amount of U.S. energy production and a large share of U.S. nuclear production, with PJM specifically having the most of any market.

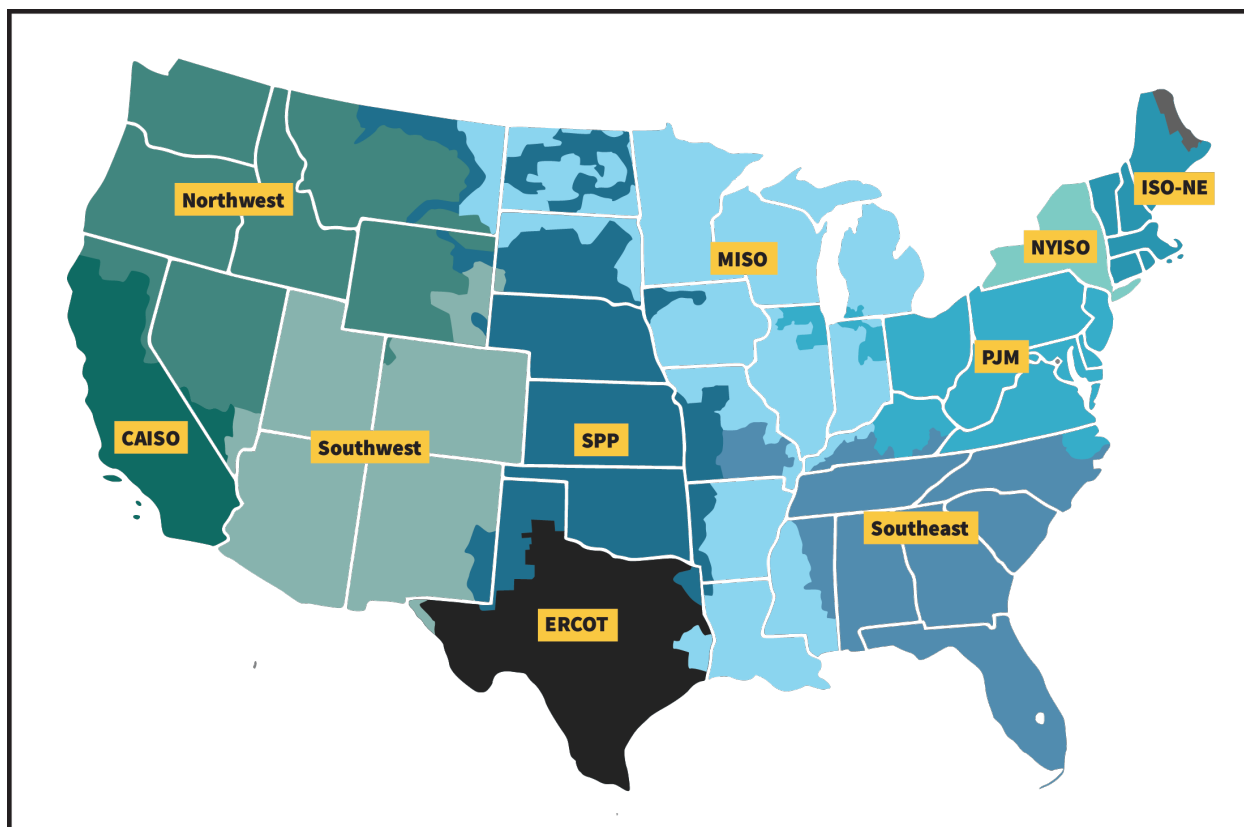


Figure 2: Geographical coverage of U.S. electricity markets

After gathering these datasets, we checked the drops in nuclear generation over each size. However, they both have wholesale electricity markets which is what matters for the purpose of this study. There are also three regions (Northwest, Southwest, and Southeast) consist of groups of independent balancing authorities (BAs), called power pools, which are groups of BAs that agree to coordinate together, to varying degrees. Since these are not unified markets, EIA does not have data on market wide generation, load or zonal pricing.

³Despite being called Southwest Power Pool, SPP is an RTO and not a power pool. The name comes from the fact that SPP used to be a power pool and more recently became an RTO.

interval, keeping only those with drops greater than 500 MWh. Smaller drops would be much more likely to be swamped by random noise in other generation sources, especially in a massive RTO like PJM. We matched each drop with a corresponding scram in the NRC database, recording the power plant, zone and time the incident occurred. This gave us a list of treatment times and zones.

We proceeded to combine the datasets together. CAISO, NYISO and SPP all had generation and price data in 5 minute intervals, and we were easily able to slice 48 hour intervals for each treatment time (24 hours on each side) and filter prices to be specific to the zone of the corresponding plant. SPP’s price data was not zonal, however, and so we instead used the South Hub price, as it is the more liquid of the two SPP pricing hubs and is preferred by traders [13]. PJM, however, only had price data at 5 minute intervals, and we had to resample and interpolate generation data from hourly up to every 5 minutes. We ended up not using any of the generation data in the regression beyond applying the drop size interaction to the treatment dummy, which came from the NRC database. Finally, we resampled and interpolated the hourly load data for all four markets. Table 1 shows the variables used, where they were used, and how they were manipulated.

Table 1: Description of Datasets and Variables

Dataset	Variable	Description	Conversion	Freq.
All	UTC Timestamp	UTC time of interval	Converted to Datetime	5 min / hourly
All	Hour Number	Hour of the day in local time	Included as fixed-effect dummies	Hourly
Prices	Zonal Marginal Price	Price of electricity in a given zone (\$/MWh)	None	5 min
Generation	Nuclear Generation	Generation from nuclear reactors (MW)	Take inter-interval differences	5 min / hourly
Load	Zonal Load	Actual load on the grid in a given zone (MW)	None	Hourly
NRC scrams	Event Time	Timestamp when a reactor scram occurred	Convert from local time into UTC and construct before/after treatment times	Irregular

Notes: Frequencies denote the native sampling interval (e.g., 5 min, hourly, or irregular).

A few scrams were dropped during this process. Two were dropped from SPP due to lack of price data, as sometimes electricity markets stop operating during periods of extreme price volatility. Additionally, three more were dropped from PJM for having missing price

and load data. Finally, one last scam was dropped because of the extreme generation and price volatility during the treatment window.

3.2 Electricity Price Panels

Once the scrams were decided on, and the combined dataset was constructed, we are able to see the price trend over time. Figure 3 shows price data plotted in the treatment window for non-PJM scrams, with treatment time marked in red. Figure 4 shows this for each PJM scam.

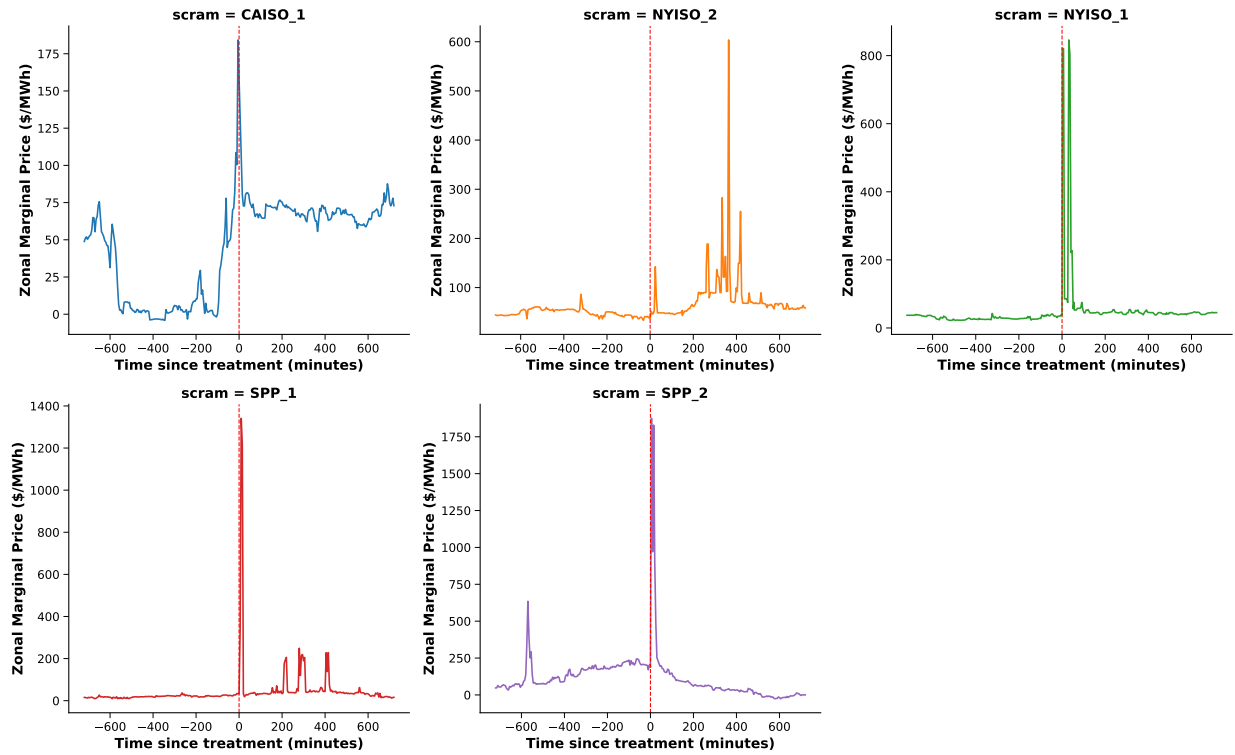


Figure 3: Zonal Prices (\$/MWh) in NYISO, CAISO and SPP over time, with treatment marked in red

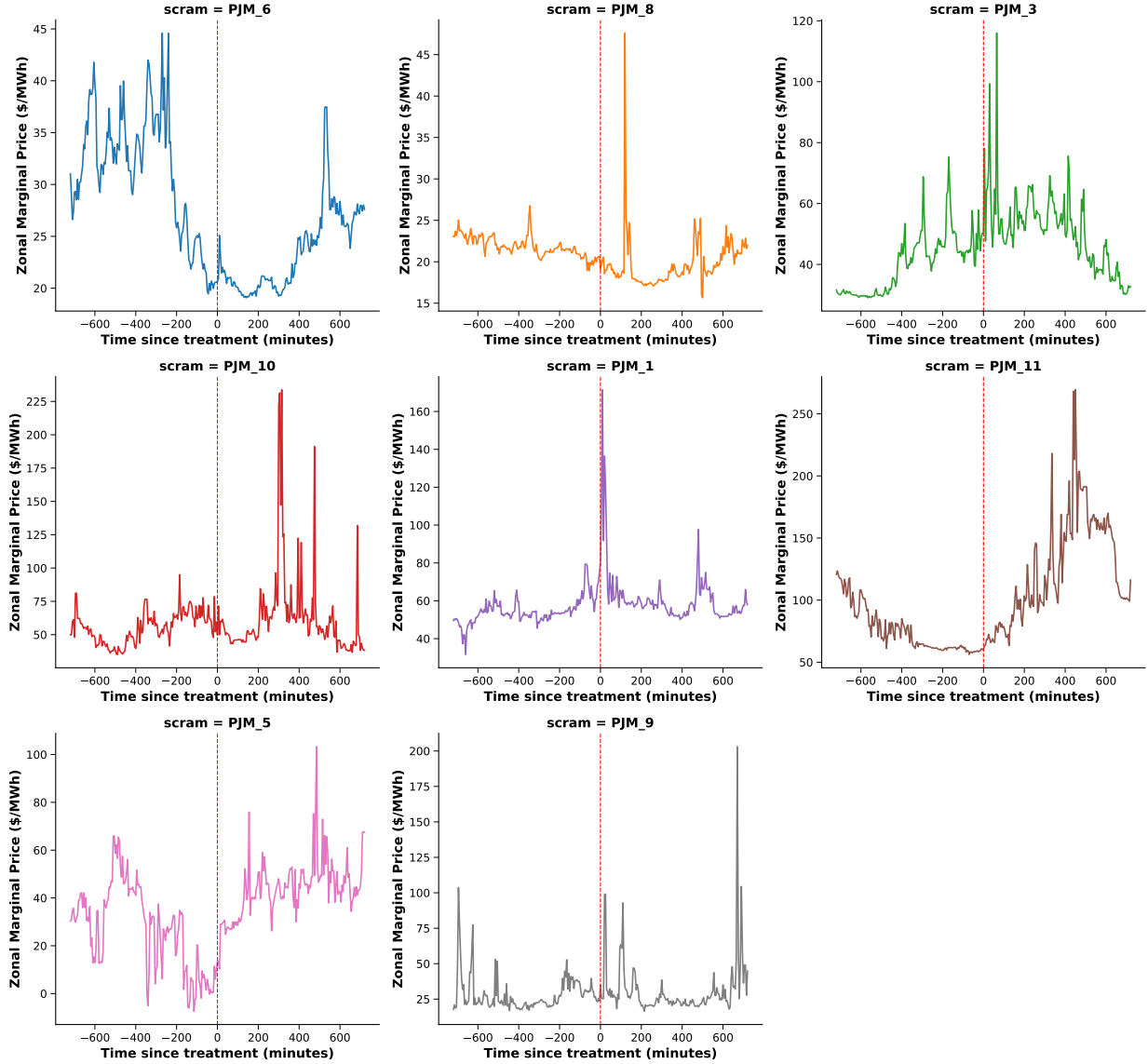


Figure 4: Zonal Prices (\$/MWh) in PJM over time, with treatment marked in red

As shown in the above figures, every scram shows at least a small spike in power prices post-treatment, with some showing massive spikes. This is a good gut-check, as the naive effect size looks to be positive. It would not necessarily be indicative of a null effect should there be no jump in prices, but the consistent existence of a jump is promising.

3.3 Panel Data Summary Statistics

Table 2 shows the means of each variable grouped by market, plant and scram. Each scram refers to an a single event of a nuclear plant going offline, except for NYISO_1, which saw a nearly simultaneous drop at Nine Mile Point and James FitzPatrick. T refers to the number of observations in time per scram. One observation is a 5-minute interval for a given scram, plant and market.

Table 2: Summary statistics by Market, Plant, and Scram (Before/After treatment)

Market	Plant	Scram	Treatment	T	MW Lost	ZMP	Load
CAISO	Diablo Canyon	CAISO_1	Before	144	–	22.098	11140.806
		CAISO_1	After	145	–988.0	69.148	10644.945
NYISO	Nine Mile Point	NYISO_2	Before	144	–	49.150	1736.374
		NYISO_2	After	155	–782.0	77.396	1644.706
	Nine Mile Point / James FitzPatrick	NYISO_1	Before	144	–	29.996	1561.708
		NYISO_1	After	153	–2117.0	74.080	1679.788
PJM	Braidwood Station	PJM_6	Before	144	–	31.191	12645.245
		PJM_6	After	145	–1153.6	23.411	10525.111
	Donald C. Cook Nuclear Plant	PJM_8	Before	144	–	21.867	14455.729
		PJM_8	After	145	–1090.9	19.700	12056.744
	Hope Creek Generating Station	PJM_9	Before	144	–	28.180	1002.558
		PJM_9	After	145	–1044.0	31.423	1059.423
	LaSalle County Station	PJM_5	Before	144	–	27.025	8511.46
		PJM_5	After	145	–1161.9	43.848	9001.464
	North Anna Power Station	PJM_11	Before	144	–	75.459	13315.353
		PJM_11	After	145	–1003.1	124.332	16250.498
	Susquehanna Steam Electric Station	PJM_1	Before	144	–	54.578	4311.670
		PJM_1	After	145	–1314.0	61.419	3885.576
		PJM_10	Before	144	–	55.360	5098.954
		PJM_10	After	145	–1013.3	62.401	5338.776
		PJM_3	Before	144	–	40.162	3815.130
		PJM_3	After	145	–1198.1	50.782	4270.923
SPP	Wolf Creek Generating Station	SPP_1	Before	144	–	19.818	3684.206
		SPP_1	After	144	–1205.7	68.784	5223.411
		SPP_2	Before	143	–	147.428	5196.965
		SPP_2	After	145	–1191.5	88.258	4776.658

Notes: The headline specification (12-hour bandwidth) did not use all of these observations. The before and after means for ZMP (Zonal Marginal Price), load are shown. ZMP is in \$/MWh, Load is in MW. T typically has an additional observation after treatment compared to before since the interval of treatment is included in after. MW lost refers to the decrease in production from the plant that scrambled. NYISO scrams with more than 144 observations are because that RTO releases more frequent data when prices spike, adding additional datapoints, but the time interval is the same. Not all PJM scrams are listed as some were dropped (see Section 3.1).

As shown in Table 2, the post-treatment mean in ZMP is typically higher than the pre-treatment mean, but not always. This is again evidence in favor of our hypothesis that

generation shocks lead to higher prices. Load is more mixed, but this is to be expected as some scrams occurred during the ramp up period of the grid in the early morning, where solar panels come online and other sources increase production.

4 Results

4.1 OLS and Main Specification Estimates

We now proceed to the analysis portion of the study. We begin with OLS of prices on nuclear generation, as well the main specification and one alternate, shown in Table 3. The main specification for this paper is as listed in (1), with fixed-effects at the scram level, a 24-hour treatment window (12 hours on each side) and clustering at the market level. The alternate specification here is the same except for fixed-effects at the plant level.

Table 3: Estimates of ρ by Specification and Fixed Effects

Specification	OLS	Scram FE	Plant FE
Load Only	-0.012805 (0.013824)	0.026176 (0.007683)	0.027106 (0.007338)
No Controls	-0.018373 (0.011784)	0.027046 (0.010441)	0.026447 (0.010460)

Notes: Each cell reports $\hat{\rho}$ with its standard error in parentheses. All specifications include time-fixed effects and are clustered at the market level. Observations: 1456 for OLS; 3,745 otherwise. OLS has less observations because PJM scrams were dropped since PJM generation data is too coarse to be useful.

As shown in Table 3, the OLS results show an effect size of $-\$0.012805 \pm 0.027095/\text{MWh}$ per MW lost, or $-\$12.81 \pm 27.10/\text{MWh}$ per GW lost. The OLS results here show the difficulty in simply running OLS to get the parameter of interest, as we get a result statistically indistinguishable from 0, and a somewhat obviously nonsense effect size, as a negative coefficient means that losing generation capacity *decreases* prices, which is highly unlikely given the laws of supply and demand.

However, our main specification returns an effect size of $\$0.026176 \pm 0.01505868/\text{MWh}$ per MW lost, or $\$26.18 \pm 15.06/\text{MWh}$ per GW lost, which is highly significant with $p = 0.0007$.

Table 3 also shows that fixed effects at the plant level instead of scram level barely changes effect sizes and standard errors, providing an initial robustness check on our results. See Section 5 for more discussion on robustness.

4.2 Comparison to Other Studies

To contextualize our effect size, it is useful to compare it to estimates from other studies. Woo et. al (2014) found that the San Onofre shutdown in California raised prices by \$6-9/MWh [10]. Since San Onofre generated 2150 MW [14], the estimated treatment effect per GW lost is \$2.79-4.19/MWh. This study does also argue that part of their effect was soaked up by reduced demand and additional supply, which could partially explain the difference.

Rinne (2019) found an effect size of 17.4 Euros per MWh or roughly \$19.74 per MWh [15], with 2.3 GWs going offline, giving an effect of \$8.60/MWh per GW lost. Given these estimates, our effect size is quite large, and when measured against prices in the data, we see estimated jumps that can sometimes more than double prices.

One possible reason for the discrepancy here is the aggregate nature of the effect size. It's possible (and likely true) that different regions have substantially different price elasticities of supply. Regions with massive generation capacities like PJM may see small effect sizes, even after accounting for zones, while regions like NYISO or SPP with comparatively little nuclear generation may see large swings. Figure 3 shows all non-PJM markets saw massive spikes immediately after treatment, while Figure 4 shows that while some of the PJM scrams translated into large spikes, some translated into a small or indiscernible spike. These effects once averaged may explain the higher than other studies estimated effects here.

However, these effects are not dramatically different than other estimates, which provides evidence that these are at least *plausible* estimates of the true size. Had we gotten results in the hundreds of dollars or less than a dollar, there would be significant reason to suspect an error here.

5 Robustness

5.1 Alternate Specifications

We now provide further evidence for the validity of our identification strategy. First, there are a variety of alternative model specifications that could be constructed using the data. The adjustments fall into three groups: entity, trend type and bandwidth. Entity refers to the level at which we apply fixed effects. The headline result is at the scam level, but we could have done this at the plant or market level⁴. Additional clustering specifications were not tested as this would only deflate standard errors and would not be useful for testing whether the effect size is real. Trend type refers to the polynomial degree of the pre- and post-trend running variables. The headline result uses a locally linear term, but we could use locally constant or locally quadratic. Lastly, the bandwidth refers to the length of time before and after treatment the window encompasses. The headline result uses 24 hours, 12 hours before and after treatment. The bandwidth could theoretically be anything, but a reasonable choice is anywhere between 2 hours and 48 hours (1 hour before/after and 24 hours before/after). Figure 5 shows a panel of all combinations of these specifications, plotted over the bandwidth.

⁴Zone level is also possible but this would be similar to plant level so it was not tested.

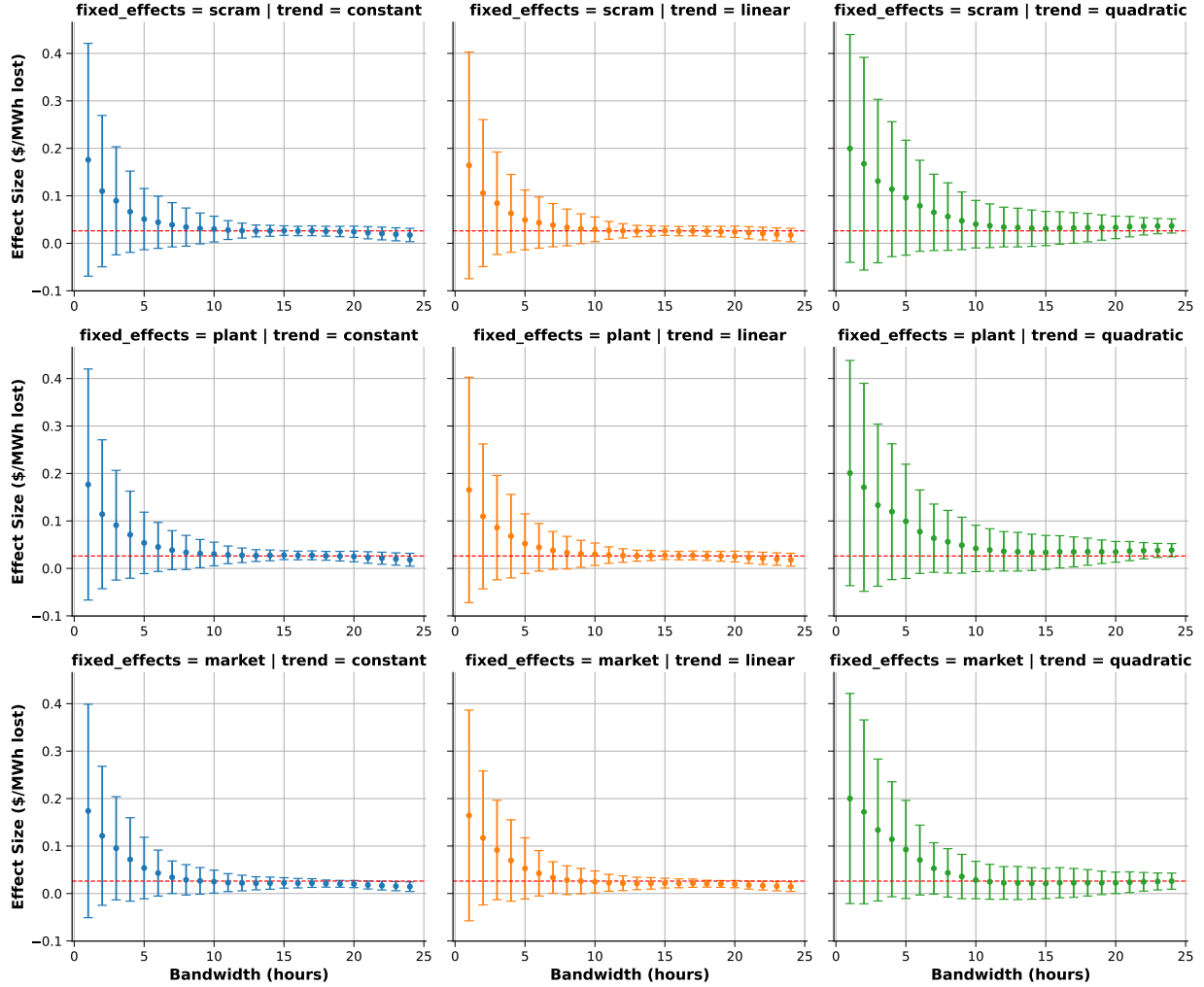


Figure 5: Effect size with confidence intervals for different specifications for fixed effects, trend and bandwidth. Headline statistic marked in red.

The headline result is within the confidence interval of nearly every specification and bandwidth level, and never far outside it, showing that the effect size is robust to a wide variety of changes. The choice of 12 hours for the bandwidth can also be justified here, as prior to it, the confidence intervals are quite large, reflective of the smaller number of observations. Larger bandwidths begin to see small declines in effect size, consistent with the introduction of bias. 12 hours is thus a reasonable middle ground that balances the bias-variance tradeoff here.

5.2 Temperature

One question that may arise is why temperature was not considered. A/C and heating are significant determinants of demand and can heavily affect the grid. Additionally, the temperature could also affect the nature of activity in the zone on that day, as on extremely hot or cold days we would expect more indoors activity and more grid demand. Table 4 our initial specifications with the addition of temperature from two major cities in each market.

Table 4: Estimates of ρ by Specification and Fixed Effects

Specification	OLS	Scram FE	Plant FE
Standard Specification	-0.012805 (0.013824)	0.026176 (0.007683)	0.027106 (0.007338)
Load & Temperature Controls	-0.015970 (0.013764)	0.027608 (0.009907)	0.026426 (0.007956)
Temperature Only	-0.018373 (0.011784)	0.031641 (0.013347)	0.026162 (0.011915)

Notes: Each cell reports $\hat{\rho}$ with its standard error in parentheses. All specifications include time-fixed effects and are clustered at the market level. Observations: 2,914 for OLS; 3,745 otherwise. OLS has less observations because PJM scrams were dropped since PJM generation data is too coarse to be useful.

We can see that while the effect sizes are essentially unchanged, the inclusion of temperature substantially inflates standard errors, indicating that this covariate does not add useful information and could be highly collinear with another predictor. To test this, Table 5 shows a Variance Inflation Factors (VIF) test on each regressor, including temperature.

Table 5: Variance Inflation Factors (VIF) for Regression Covariates

Variable	VIF
mc2_temp	67.389
mc1_temp	60.603
treatment	6.294
actual_load	4.029
hour_number	3.728
after_treatment	3.707
before_treatment	3.149

Notes: VIFs calculated for each covariate in the main regression. High VIFs (e.g., for mc1_temp and mc2_temp) indicate strong multicollinearity among temperature variables.

We can see that the temperature covariates have VIFs of over 60, indicating a catas-

trophic collinearity problem, as typically above 10 is considered an issue. This makes intuitive sense, as load should be sufficient to capture variation in temperature across time, as the rationale for including temperature is precisely that it affects load. Additionally, any residual variation not explained by load would probably be captured by hour of day fixed effects, as temperature is relatively variable along that dimension.

5.3 Falsification Tests

The final way we test robustness is to artificially shift the treatment date and conduct a 'placebo' regression. If the effect is real, we would expect to see estimates of zero or near zero for all shifts that are not 0 (the baseline model). Figure 6 shows the effect size against the treatment time shift. Besides the shifted treatment, everything else is the same as the baseline model.

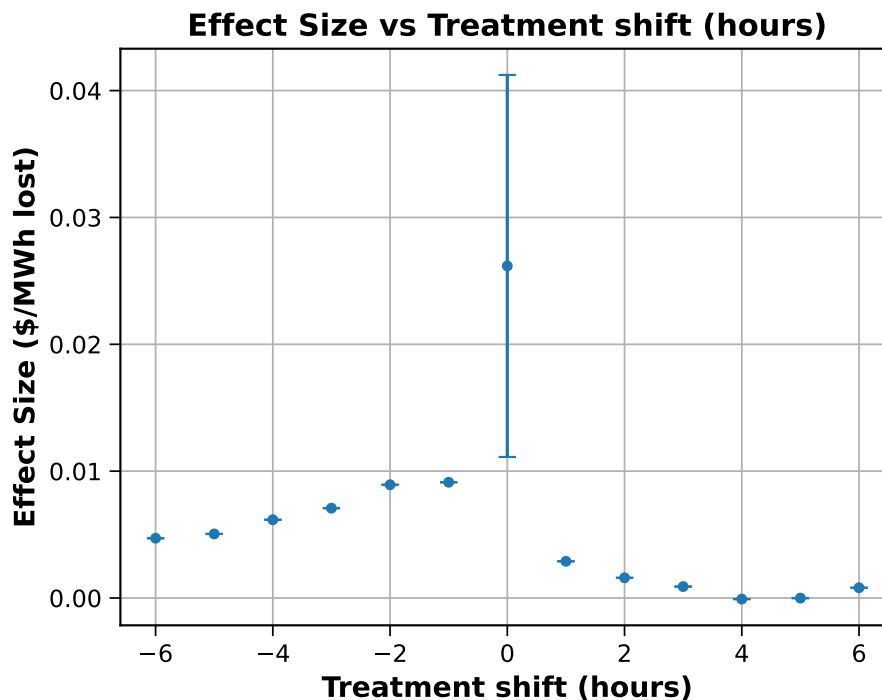


Figure 6: Effect size with confidence interval versus time shift of treatment

While most of the placebos are not exactly zero, all of them are pretty close. More

importantly, they are all substantially smaller than the non-shifted treatment and outside the confidence interval of the true effect size. Lastly, the extremely small confidence intervals for the shifted treatments indicate that the model is fitting on the underlying smooth trend without the discontinuity, which has little variation, showing that there really is an effect at the treatment times. This is all evidence in favor of the effect being real, and provides additional strength to the validity of the estimated causal effect.

6 Conclusion

Nuclear energy has played a vital role in the global energy system for decades, but its future is under continued threat. However, there has recently been momentum to save these reactors, and to once again expand the nuclear industry. With modern designs like small modular reactors being developed, there is significant opportunity to utilize nuclear energy’s potential. Our study aims to provide evidence to engineers, policymakers and regulators in favor of nuclear energy, and highlight the significant costs in continuing down the path of closure. By exploiting instantaneous shocks in the form of nuclear scrams via an RDiT approach, We find that loss of nuclear generation can cause large price hikes, possibly magnified in smaller areas. Despite the analyses conducted here, there is still substantial room to conduct further analyses. For starters, work could be done on understanding the heterogenous effects of each shock, and what drives that heterogeneity. Additional scrams, plants and regions could and should be included to better understand the effect of lost generation more broadly. Finally, more work also can be done on decomposing the effect size found here into what channels the shock propagates through.

In conclusion, our study enriches the existing literature on nuclear closure-driven price hikes, and provides additional insight into short-run price dynamics.

7 Appendix

7.1 Main Specification Full Regression Results

Table 6: PanelOLS Estimation Summary

Dep. Variable: lmp Estimator: PanelOLS Cov. Estimator: Clustered Observations: 3768 R²: 0.0888 (Between: -2.3439; Within: 0.0865; Overall: -0.9504) Entities: 13 (Min: 286; Max: 298; Avg: 289.85) Time Periods: 25 (Min: 57; Max: 161; Avg: 150.72) F-statistic: 90.754 (df=4, 3727; $p < 0.001$) Robust F: 2.238e+15 (df=4, 3727; $p < 0.001$)						
Parameter	Estimate	Std. Err.	T-stat	P-value	Lower CI	Upper CI
treatment	0.0262	0.0077	3.4069	0.0007	0.0111	0.0412
before_treatment	0.0222	0.0133	1.6693	0.0951	-0.0039	0.0483
after_treatment	-0.0682	0.0485	-1.4047	0.1602	-0.1633	0.0270
actual_load	0.0184	0.0070	2.6174	0.0089	0.0046	0.0322
F-test for Poolability: 27.334 (df=36, 3727; $p < 0.001$) Included effects: Entity, Time						

References

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